
Problems for Ch 4 RW – Motion in 2 and 3 Dimensions
 Addisionele Probleme vir Hfst 4HRW – Beweging in 2 of 3 dimensies

	9 th Edition	8 th Edition
Questions	5, 7, 9, 11, 12	4, 9, 7, 13, 11
Problems	1, 3, 7, 11, 27, 32, 33, 35, 58, 80, 82, 108	3, 1, 5, 13, 27, 38, 31, 39, 62, 78, 82, 92

1	3	6.2 m
3	1	$(-2.0\text{m})\hat{i} + (6.0\text{m})\hat{j} - (10\text{m})\hat{k}$
7	5	$(-0.70\text{m/s})\hat{i} + (14.0\text{m/s})\hat{j} - (0.40\text{m/s})\hat{k}$
11	13	(a) $(6.00\text{m})\hat{i} - (106\text{m})\hat{j}$ (b) $(19.00\text{m/s})\hat{i} - (224\text{m/s})\hat{j}$ (c) $(24.0\text{m/s}^2)\hat{i} - (336\text{m/s}^2)\hat{j}$ (d) -85.2°
27	27	(a) 10.0 s (b) 897 m
32	38	(a) 12 m (b) 19.2 m/s (c) 4.8 m/s (d) Since $v_y > 0$ when the ball hits the wall, it has not reached the highest point yet.
35	39	4.84 cm
58	62	(a) 0.94 m (b) 17.0 m/s (c) $1.9 \times 10^3 \text{ m/s}^2$ (d) 55 ms
80	78	(a) 7.1 m/s (b) 102° (measured from the $+x$ axis), or 16° west of north (c) 29 s NOT 28 s
82	82	(a) 34° west of north (b) 48.3 s
108	92	(a) $6.1 \times 10^3 \text{ m}$ (b) 24 m/s